# UPI SHIELD AI

Explainable AI-Driven Scam-Pattern Recognition

**Team RSG** • Muthayammal Engineering College

Hackathon track: AI-Driven Scam Pattern Recognition

Synthetic-data prototype combining transparent rules, Isolation Forest anomaly signals, explainable risk indicators, and a human-review workflow.

SILAMBARASAN C. • RAMANA • GOKUL PRIYA

Prototype only — not connected to real UPI systems and does not initiate or block payments.

# 1. The problem

**Digital payments are fast, but suspicious patterns can be hard to review consistently.**

Potential signals include unusually large amounts compared with typical activity, rapid transaction bursts, and new-payee activity. Any one signal can also occur during legitimate use.

The goal is to prioritize review without treating an anomaly as proof of fraud.

# 2. Our proposed solution

**UPI Shield AI** combines transparent detection rules with a machine-learning anomaly signal.

- Rule-based checks provide readable reasons for an alert.
- Isolation Forest highlights unusual feature combinations.
- Risk indicators prioritize cases for human review.
- A dashboard and reports support inspection and evaluation.

Design principle: explain the signal, preserve uncertainty, and keep a human in the loop.

# 3. Prototype workflow

Synthetic transaction events → feature engineering → rule-based detector + Isolation Forest → risk assessment → explanation → analyst review → evaluation report.

Demo inputs include transaction amount, typical amount, recent transaction count, and new-payee status.

No live banking data, UPI integration, payment initiation, or automatic blocking is implemented.

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# 4. Detection design

**Rule-based layer:** evaluates predefined patterns such as amount deviation and transaction frequency, then provides interpretable reasons.

**ML layer:** Isolation Forest identifies unusual feature combinations. Its output is an anomaly signal, not a calibrated probability of fraud.

**Hybrid review:** combines rule evidence and ML signal to prioritize analyst attention. New-payee status alone should not produce a high-risk conclusion.

Real deployment would require authorized representative data, independent validation, monitoring, and security review.

# 5. Synthetic evaluation results

Held-out synthetic test set: **90 events** — 71 actual normal and 19 actual suspicious.

| Metric | Rule-based | Isolation Forest |
|---|---|---|
| Precision | 100.0% | 73.7% |
| Recall | 100.0% | 100.0% |
| F1-score | 100.0% | 84.8% |

**Isolation Forest confusion matrix:** TN = 71, FP = 5, FN = 0, TP = 14.

These results apply only to this synthetic test set. The rule-based 100% scores may reflect how the synthetic labels and rules were constructed; they do not establish real-world fraud detection performance.

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# 6. Hybrid scenario validation

**Observed scenario-test results:** all available hybrid checks passed in the current notebook run.

- Normal activity: score-range check passed; LOW risk returned.
- New payee only: score-range check passed; rule risk remained below HIGH.
- High amount + transaction burst: score-range check passed; HIGH risk returned.

These are targeted scenario checks, not comprehensive security testing. Expand coverage with edge cases, malformed inputs, boundary values, and regression tests.

# 7. What the metrics mean

- Isolation Forest flagged all 14 synthetic suspicious events but also flagged 5 normal events.
- Precision of 73.7% means some flagged events were false alarms in this evaluation.
- Rule-based metrics are perfect on this particular synthetic set; independent and more realistic testing is necessary.
- Future work: improve scenario diversity, tune thresholds on validation data, test robustness, and check for data leakage.

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# 8. Privacy, security and safety

- Synthetic demonstration data only; do not enter payment PINs, passwords, or real customer records.
- No payment initiation, verification, or automatic blocking.
- An anomaly is not proof of fraud; human review is required.

Before deployment: data authorization and minimization, access controls, secure audit logs, abuse testing, drift monitoring, and independent evaluation.

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# 9. Team RSG

**SILAMBARASAN C.** — suggested role: Team Lead & AI/ML (dataset, detection logic, model evaluation, coordination).

**RAMANA** — suggested role: Prototype & Dashboard (interface, risk display, explanations, review queue).

**GOKUL PRIYA** — suggested role: Research, Security & Presentation (threat model, test evidence, documentation, demo preparation).

These are proposed assignments. Confirm and edit them to match each member's actual contribution before submission.

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# 10. Next steps

1. Re-run all notebook cells from top to bottom and confirm reproducible outputs.
2. Add edge-case and regression tests; investigate any failed test.
3. Capture genuine dashboard screenshots for normal and suspicious scenarios.
4. Document synthetic data generation, train/test split, and limitations.
5. Follow the official hackathon submission rules, format, and deadline.

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# Closing

Explainable alerts for better-informed human review.

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Synthetic-data prototype. Results are preliminary and do not represent real-world UPI fraud detection performance.